

Chedam: work log

Cumulative record of every working day, newest first. This copy covers up to Monday, October 5, 2026. Owner: Sreya.
Baseline: Master Specification v1.0.

Days in this log

- **2026-10-05:** P0 steps 1-3 done: hub base on the new network, schema + event log, access (roles, PINs, lockout); work log started
- **2026-10-03:** New network: own router, Pi re-flashed, SSH keys documented, HDMI fixed
- **2026-10-02:** P0 kickoff: spec review, dev PC tools, first Pi flash and first hub setup (TELUS network)

P0 steps 1-3 done: hub base on the new network, schema + event log, access (roles, PINs, lockout); work log started

Summary

- Full health check of the re-flashed Pi: healthy, OS up to date, no hub software yet.
- Restored admin access (the sudoers rule is lost on every re-flash).
- Changed `setup-hub.sh` so the display setting can be switched **both ways** (monitor on or headless) and re-runs leave the boot config unchanged.
- Ran hub setup with `--keep-display`, rebooted and verified everything. HTTPS works by name and by IP.
- iPhone 13 Pro and Android 9 installed the new "Chedam Store CA" and opened the hub page without warnings ("Chedam hub is running").
- Started this cumulative work log (`Work Log/`).
- P0 step 1 (hub base) is complete.
- P0 step 3 (access) is complete: permission engine with role templates, overrides and BR-33, PIN sign-in with lockout, owner recovery code, "Who can do this?". 92 more tests (124 in total) passing; deployed and checked on the Pi.
- P0 step 2 (schema v1 + event log) is complete: 14 tables, reference data, dev sample store, a transactional append-only event log, 32 automated tests passing, deployed to the Pi with a backup first.

Naming note: the local folder `chedam-updates/` in the project is Sreya's YouTube and notes repo (gitignored). It is **not** the update channel. Nothing is renamed; the public update-package repo name will be decided when step 9 (Updates) starts.

1. Health check (before any change)

One SSH session ran these checks:

```
whoami; hostname; hostname -I; cat /proc/device-tree/model; grep PRETTY /etc/os-release; uname -r
uptime; sudo -n true; free -m; df -h / /boot/firmware; lsblk
vcgencmd measure_temp; vcgencmd get_throttled; vcgencmd measure_volts core
timedatectl; ip route; ping -c2 1.1.1.1
apt list --upgradable; dpkg-query -W <packages>; systemctl is-active ...; systemctl --failed
cat /boot/firmware/config.txt /boot/firmware/cmdline.txt
```

Area	Result
Hardware	Pi Zero 2 W Rev 1.0, 33.2 °C, throttled 0x0, core 1.2875 V
OS	Debian 13 trixie, kernel 6.18.50+rpt-rpi-v8, 0 pending updates
Uptime	1 day 12 h, load 0.00, no failed units
Memory / disk	415 MB total, 273 MB available / 119 GB card, 2.3 GB used
Network	192.168.50.101 via 192.168.50.1, internet ping 7 ms
Time	NTP synced, America/Vancouver, no RTC
sudo	Needed a password (re-flash reset it)
Hub software	Not installed: no caddy, chrony, ufw, sqlite3 or <code>/opt/chedam</code>
Display	<code>video=HDMI-A-1:1280x720@60D</code> present (monitor in use)

2. Admin access

Failed case: `Permission denied (publickey)` when restoring sudo

- The `ssh -t chedam "...sudo tee..."` command was typed **inside an SSH session on the Pi** (prompt `chedam@chedam:~ $`).
- So the Pi tried to SSH into itself and had no key for that. Re-flashing was not needed.
- Fix: run only the inner command on the Pi:

```
echo 'chedam ALL=(ALL) NOPASSWD:ALL' | sudo tee /etc/sudoers.d/010-chedam && sudo chmod 440 /etc/sudo
sudo -n true && echo OK
```

3. Display becomes a switch (DL-24 updated)

Sreya's requirement: the monitor is optional on the Zero 2 W pilot, but the option must stay. Pi 4 production hubs will use a display to operate the hub directly.

Change in `hub/scripts/setup-hub.sh` (boot config section):

- Removes any earlier chedam block, then writes a block between `# chedam-begin` and `# chedam-end`.
- `--keep-display`: display driver `vc4-kms-v3d` and display detection **on**, Bluetooth off.
- No flag: headless. The display driver and detection are off, `gpu_mem=16`, Bluetooth off; this saves ~50 MB of RAM.
- Audio and camera detection are always off.

```
sudo bash ~/hub/scripts/setup-hub.sh --keep-display # monitor on (current)
sudo bash ~/hub/scripts/setup-hub.sh # headless
sudo reboot # after switching
```

Failed case: an extra `[a11]` line on every run

- The test (switching off, on, off, on on a copy of `config.txt`) worked, but each run appended another `[a11]` line. It was harmless but untidy.
- Fix: moved `[a11]` inside the marked block. The original config (`config.txt.chedam-orig`, saved by the first run) was restored and setup run twice. Both runs gave the same checksum (`d91ff929`), and the only differences from the original were audio off, camera detection off and the chedam block.

4. Hub setup on 192.168.50.101

```
scp -r scripts system chedam:hub/
# on the Pi: download + verify PocketBase
wget pocketbase_0.40.4_linux_arm64.zip checksums.txt
grep linux_arm64 checksums.txt | sha256sum -c - # OK
sudo apt-get update
sudo bash ~/hub/scripts/setup-hub.sh --keep-display
sudo systemctl reboot
```

Verification after reboot:

Check	Result
Services	chedam-hub, caddy, chrony, avahi-daemon, ufw active; no failed units
Memory	PocketBase 28 MB + Caddy 37 MB = 65 MB (budget 150 MB, NFR-08); 264 MB available with display on
Watchdog	RuntimeWatchdogUsec=14s
Time	chrony, 0.0000007 s from NTP, leap normal
Temperature	38.6 °C, throttled 0x0
Display	/dev/dri/card0 present
Firewall	22, 80, 443/tcp and 5353/udp allowed (IPv4 + IPv6)
CA download	<code>http://chedam.local/ca.crt</code> → 200, "Chedam Store CA - 2026 ECC Root", valid to 2036-08-13
HTTPS	<code>https://chedam.local/api/health</code> and <code>https://192.168.50.101/api/health</code> → 200
Certificate check	<code>openssl s_client ... -CAfile ca.crt</code> → Verify return code 0, with and without SNI

HTTP	Redirects to HTTPS (302)
Caddy errors since boot	0

PC test commands (Git Bash; `--resolve` avoids Windows mDNS problems, and `--ssl-no-revoke` avoids the Schannel revocation error):

```
curl --resolve chedam.local:80:192.168.50.101 -o ca.crt http://chedam.local/ca.crt
curl --ssl-no-revoke --resolve chedam.local:443:192.168.50.101 --cacert ca.crt https://chedam.local/a
curl --ssl-no-revoke --cacert ca.crt https://192.168.50.101/api/health
```

5. Phones (risk R5, first part)

Device	How	Result
iPhone 13 Pro, iOS 26.6.1	Safari <code>http://chedam.local/ca.crt</code> → install profile → Settings > General > About > Certificate Trust Settings → on	<code>https://chedam.local</code> shows the placeholder, no warning
Android 9	Chrome <code>http://192.168.50.101/ca.crt</code> → install for VPN and apps	<code>https://192.168.50.101</code> shows the placeholder, no warning

The camera test over HTTPS comes later, when the client app exists.

6. Work log started

- Folder `Work Log/`: one Markdown source per day in `Work Log/entries/YYYY-MM-DD.md`.
- `tools/worklog/build_worklog.py` builds `Work Log/Chedam-Work-Log-YYYY-MM-DD.pdf` with **all** days, newest first.
- Earlier PDFs are kept, so each day has its own complete snapshot.

```
python tools/worklog/build_worklog.py          # newest day
python tools/worklog/build_worklog.py 2026-10-05 # a specific day
```

The days 2026-10-02 and 2026-10-03 were written up from the session records and `docs/P0-pi-setup-log-2026-10-03.pdf`.

7. P0 step 2: schema v1 and event log

Spec basis: Section 10 (data model, event log), Section 13 (build process: migrations and sample data first), NFR-20 (reversible migrations), BR-30 (UTC), BR-33 (managers cannot grant more than they hold).

Files added

File	Purpose
<code>hub/pb_migrations/1791200001_p0_schema_v1.js</code>	13 new tables + the PocketBase <code>users</code> collection extended: business, locations, settings, modules, permissions, roles, permission_overrides, devices, storage_areas, tasks, events, backups, updates. Common fields on every table. API rules all closed (superuser only) until step 3
<code>hub/pb_migrations/1791200002_p0_reference_data.js</code>	22 modules (3 core always on, 15 switchable, 4 later), shop-type presets, 20 P0 permissions (5 owner-only), 5 role templates, 12 default settings
<code>hub/pb_migrations_dev/1791200100_dev_sample_store.js</code>	Dev-only sample store: "Demo Grocery", main location, grocery preset modules, 4 storage areas, 5 people (one per role, random unknown passwords), 1 setup task
<code>hub/pb_hooks/event_log.pb.js</code> + <code>lib/event_log.js</code>	Event log: one <code>events</code> row per create/update/delete, in the same transaction; actor and device stamped from the request; append-only
<code>hub/tests/step2-schema-eventlog.test.mjs</code>	32 automated tests against a throwaway local PocketBase

hub/scripts/deploy.sh	PC → hub deploy: copy, SQLite online backup + integrity check, rsync, restart, health, migrations, errors
docs/conventions.md	Data, event-log, migration and workflow rules for all phases

Design choices

- **Common fields:** `id` (a device may create it offline), `created_at`, `updated_at`, `created_by`, `updated_by`, `device_id`, `deleted_at`. `updated_by` was added beyond the spec's list, so each change records who made it.
- **The event log runs inside the same transaction** as the change (`onRecord*Execute + runInTransaction`). If the event can't be written, the change is rolled back, so the log never misses a write.
- **Actor and device are stamped by the hub** in request hooks. A client cannot forge `created_by`, and the device comes from the `X-Chedam-Device` header (verified in step 4).
- **Hidden fields** (`pin_hash`, `recovery_code_hash`, `key_hash`, `pairing_code_hash`) never reach the API or the log.
- **Role levels:** owner 100, manager 80, accountant 50, cashier 40, staff 30. Owner-only permissions: `modules.manage`, `setup.run`, `owner.manage`, `backups.restore`, `updates.install`.
- **Sample data is separate** from real migrations, so a store hub never receives demo records.

Commands

```
# local tests (PC, Git Bash)
PB_BIN='C:\Tools\pocketbase.exe' node hub/tests/step2-schema-eventlog.test.mjs
# deploy to the Pi with sample data
bash hub/scripts/deploy.sh --sample-data
# verify on the Pi
ssh chedam 'sudo -u chedam-hub sqlite3 /opt/chedam/pb_data/data.db "select table_name, action, count ('
```

Failed cases and fixes

Symptom	Cause	Fix
pocketbase.exe: command not found in Git Bash	That shell started before <code>C:\Tools</code> was added to PATH	Full path <code>C:\Tools\pocketbase.exe / PB_BIN</code>
migrate down hung	It waits for a y/N confirmation	Pipe <code>y</code> (input: <code>"y\n"</code> in the tests)
Python check query error (ESCAPE expression must be a single character)	Backslash escaping in an inline script	Filter with <code>substr(name,1,1) != '_'</code>
Sample migration: Cannot read property 'forEach' of undefined	A JSON field read with <code>get()</code> returns raw bytes	<code>JSON.parse(record.getString("value"))</code>
Test stop step hung	Waited for the exit of a process that had already exited	Check <code>exitCode/signalCode</code> first
Create returned 400 "Something went wrong"	<code>e.request</code> doesn't exist in PocketBase 0.40 JS hooks (found with <code>serve --dev logs</code>)	<code>e.requestInfo().headers["x_chedam_device"]</code> (names lowercased, <code>_</code> → <code>_</code>)
A Python edit put real newlines inside JS strings	Escaping in a heredoc	Fixed with a direct edit

Test results (PC, 32/32 passed)

- Schema/reference: 22 modules, 3 core on, 15 switchable, 20 permissions, 5 roles, owner holds all, manager holds no owner-only permission, 12 settings.
- Sample: business + location, 4 storage areas, 5 people, 1 task, 7 grocery modules on.
- Event log: device-made id accepted; forged `created_by` overwritten; device stamped; create/update/soft-delete/delete each logged with correct before/after/changed; delete keeps the last state; no PIN hash, password or

tokenKey in the log; events can't be edited or deleted, even by a superuser.

- Atomicity: with the event write forced to fail, the create fails and the record is not in the table.
- Migrations: down 3 and up again both clean.

On the Pi (after deploy)

Check	Result
Pre-deploy backup	pre-deploy-<UTC>.db, integrity ok
Migrations applied	schema v1, reference data, dev sample store
Rows	modules 22, permissions 20, roles 5, settings 12, business 1, locations 1, storage_areas 4, users 5, tasks 1
Events	78 = every write above (incl. 7 module updates for the grocery preset)
Secrets in events	0
SQLite journal	WAL
PocketBase memory	42 MB (budget 150 MB)
Errors in the journal	none

8. P0 step 3: Access

Spec basis: FR-1.03 (owner password, PIN, printed recovery code), FR-1.04 (people with role and PIN), FR-1.09 (owner signs in on any device), FR-1.10 (role templates, per-person overrides with end date, "Who can do this?"), NFR-11 (hub-side permission checks, PIN lockout after 5 tries, auto-lock), BR-33 (managers can't grant more than they hold).

Files added or changed

File	Purpose
hub/pb_migrations/1791200003_p0_access.js	PIN and recovery code become bcrypt password fields (hidden); pin_set and recovery_code_created_at added; API rules set on all 14 tables; token lifetime 12 h. Reversible
hub/pb_hooks/lib/access.js	Permission engine: effective permissions (role + allow overrides – deny overrides; owner = all), the per-table access map, protected fields, self/assignee field limits, BR-33 checks for people, roles, overrides, modules, settings, business
hub/pb_hooks/lib/auth.js	PIN rules, lockout counter, recovery code generator, secret check
hub/pb_hooks/access.pb.js	Guard on every list/view/create/update/delete; password sign-in lockout; token refresh check; endpoints (below)
hub/pb_migrations_dev/1791200101_dev_sample_pins.js	Dev-only PINs for the 5 sample people (test values; dev hubs only)
hub/pb_hooks/lib/event_log.js	Fix: put the original app back after the transaction (see failed cases)
hub/tests/lib/hub.mjs, step3-access.test.mjs, run-all.mjs	Shared test harness, 92 access tests, one-command runner
docs/conventions.md, docs/P0-mini-spec.md	Access rules for later phases; decisions DL-28 to DL-32

Endpoints

Method and path	Who	What
GET /api/chedam/auth/pin-users	anyone on the LAN (step 4: paired devices only)	Names for the till sign-in screen
POST /api/chedam/auth/pin	anyone	{user, pin} → sign-in token; wrong tries count toward lockout
POST /api/chedam/users/{id}/pin	yourself, or users.manage above them	Set or change a PIN (4-6 digits, no repeats or straight runs)

POST /api/chedam/users/{id}/unlock	users.manage above them	Clear a lockout
POST /api/chedam/owner/recovery-code	owner	New printable code, shown once (format XXXXX-XXXXX-XXXXX-XXXXX)
POST /api/chedam/auth/recover	anyone with the code	{email, code, new_password} → new password + sign-in; the code is used up
GET /api/chedam/access/me	signed in	My role and effective permissions
GET /api/chedam/access/users/{id}	self or users.view	Someone's effective permissions and where each comes from
GET /api/chedam/access/who-can/{code}	users.view	"Who can do this?" (via role or override, with end date)

Who may do what (summary of the access map)

- **Anyone signed in:** reads the business profile, locations, settings, modules, permissions, roles and storage areas.
- **People:** list with users.view; a person can view and edit their own record (name, email, phone, language, accessibility, password only); add/edit others needs users.manage; giving or changing roles needs access.manage; no hard delete.
- **Overrides:** access.manage only; never on yourself; only on people below you; never grant what you don't hold; owner-only permissions only by the owner.
- **Roles:** templates can't be deleted; only the owner edits templates; the owner role always holds everything; managers create and edit only roles below their own level.
- **Modules:** owner only; only `enabled` can change; core can't be switched off; "later" modules can't be switched on; who/when stamped.
- **Settings:** `security.*` owner only; keys can't be renamed.
- **Audit log:** events.view. Tasks: tasks.view / tasks.manage (the assignee may update status/note). Devices: devices.view / devices.manage. Backups and updates: read-only through the API.

Failed cases and fixes

Symptom	Cause	Fix
No way to check a bcrypt PIN in JS hooks	<code>\$security</code> has no bcrypt	Use a PocketBase <code>password</code> field: <code>record.getRaw("pin").validate(plain)</code>
Owner recovery sign-in failed: failed to delete all previous MFAs ... transaction has already been committed	The event-log wrapper replaced <code>e.app</code> with the transaction and never restored it, so PocketBase's clean-up after a password change used a finished transaction. This would have broken every password change.	Restore <code>e.app</code> in a <code>finally</code> block after <code>runInTransaction</code>
5th wrong password returned 400, not 423	The failure was counted, but that same response didn't report the new lock	After counting, re-read the person and answer "locked" (same as PIN sign-in)
Assumed PocketBase leaves <code>e.record</code> empty on a wrong password	Traced with debug lines: the hook gets the person and <code>e.next()</code> checks the password	Removed the extra lookup and the debug lines
Step 2 test still sent <code>pin_hash</code> and reversed only 3 migrations	The schema changed in step 3	Test uses the <code>pin</code> field, checks for no <code>\$2a\$</code> hashes, and reverses every Chedam migration it finds
<code>sed</code> edits to the test harness didn't apply	Escaping in shell quoting	Direct edits

Test results (PC)

node hub/tests/run-all.mjs → step2 32/32, step3 92/92.

Step 3 covers: - PIN sign-in, the PIN list showing no secrets, and wrong PINs refused. - Lockout after 5 wrong PINs and after 5 wrong passwords (423), and unlock by a manager but not a cashier. - The PIN rules. - Who can read and change

each table. - Managers adding cashiers but not owners or other managers, and editing people below them but not the owner. - Role changes, and the rule that you can't change your own role. - Suspending someone signs them out at once. - Overrides: allow, deny, past and future end dates, owner-only, and "can't grant what you don't hold". - Role templates, the owner role, and deleting custom roles. - The module, setting and business rules. - The recovery code: shown once, case-insensitive, used once. - Who-can and my-permissions. - The audit trail: lockouts, overrides and module switches are logged, and no hashes appear anywhere.

On the Pi (after deploy, checked on the Pi itself)

Check	Result
Deploy	Backup + integrity ok; migrations 1791200003_p0_access.js and 1791200101_dev_sample_pins.js applied; no errors
PIN list	5 people
Signed-out request	401
PIN sign-in time	0.50 s (bcrypt cost 10 on the Pi Zero 2 W)
access/me for the cashier	role cashier, permissions tasks.view
Cashier lists people	403
bcrypt hashes in the event log	0
PocketBase memory	44 MB

The dev PINs are in the dev seed file only and are not repeated in this log.

Next

P0 step 4, Devices (FR-1.07, 1.08): pairing by code/QR, certificate install guide inside the pairing flow (iPhone trust toggle, Android user CA, IP fallback for Android 9), device keys checked on every request instead of trusting the x-Chedam-Device header, device manager (online status, user, version; lock, log out, rename, assign, revoke, approve), and the PIN name list limited to paired devices.

2026-10-03 (Saturday, October 3, 2026)

New network: own router, Pi re-flashed, SSH keys documented, HDMI fixed

Source: docs/P0-pi-setup-log-2026-10-03.pdf (detailed log of this day) and docs/P0-restart-new-network.md (the plan). This entry carries its content forward.

Summary

- The TELUS modem gives no admin access, so DHCP reservations are impossible. The dev hub moved behind our **own TP-Link router**, which is LAN-cabled to the bedroom wall port.
- Network: subnet 192.168.50.0/24, gateway 192.168.50.1, 2.4 GHz SSID TP-LINK_2CBE.
- Pi chedam at 192.168.50.101 (MAC 88:A2:9E:5C:17:65, reservation enabled); dev PC DESKTOP-AHG4TD7 at 192.168.50.100.
- The SD card was wiped and re-flashed. Key-only SSH works, and the OS is updated.
- HDMI console made visible. setup-hub.sh gained the --keep-display option.
- Commits: ea0a7e6, 678eee4, bacf585.

Why SSH keys instead of a password

With a key pair, the private key never leaves the PC and the Pi holds only the public key. No secret crosses the network, keys cannot be guessed, and scripts can log in, so password login is switched off.

File	Where	Rule
id_ed25519	PC C:\Users\Venkata\.ssh\	Private. Never share, paste or commit; back it up privately
id_ed25519.pub	PC, same folder	Public, one line ending chedam-dev. Safe to copy
authorized_keys	Pi /home/chedam/.ssh/	Allowed public keys; folder 700, file 600
known_hosts	PC .ssh\	Remembered Pi identities; ssh-keygen -R <ip> after a re-flash
config	PC .ssh\	Alias ssh chedam

Key procedures recorded:

- create the key pair
- put the public key on the Pi with Imager (REMOVE old keys, then BROWSE to the .pub file)
- add or replace keys with type new.pub | ssh chedam "cat >> ~/.ssh/authorized_keys"
- what to do when locked out (monitor + keyboard, or re-flash)

Steps

1. Backed up .ssh with Copy-Item \$env:USERPROFILE\.ssh \$env:USERPROFILE\.ssh_backup -Recurse.
2. Cleared old host fingerprints by deleting known_hosts and known_hosts.old. There was a backup.
3. Router: changed the admin password, set the LAN to 192.168.50.1/24 (different from TELUS 192.168.1.x), kept it in router mode, and turned on 2.4 GHz.
4. Re-flashed the card. Imager: Pi Zero 2 W, Lite 64-bit, chedam/chedam, new Wi-Fi, CA, SSH public-key only.
5. Found the Pi in the router's DHCP client list and created the reservation.
6. Updated the SSH config to HostName 192.168.50.101 and removed HostKeyAlias.
7. Ran sudo apt update && sudo apt full-upgrade -y and checked that sshd -T shows passwordauthentication no.
8. Readiness check: model, OS, IP, active Wi-Fi, multi-user.target; reboot test passed.
9. Forced HDMI output (see below).

Failed cases and fixes

Attempt / symptom	Cause	Fix
ssh-keygen -R <old-ip> errored	PowerShell treats < as an operator	Use real addresses; deleting known_hosts avoided needing the old IP
Key pair deleted during "start fresh"	Not needed: a key is not tied to a network	Restored from .ssh_backup
ssh 192.168.50.101 denied as venkata	No user given, so the Windows name was used	ssh chedam@192.168.50.101
ssh chedam@chedam timed out	Windows cannot resolve a bare name	IP, chedam.local, or the alias
-i \$env:venkata\.ssh\... not found	\$env:USERPROFILE is a real variable, not a placeholder	Keep \$env:USERPROFILE
Permission denied (publickey)	Imager wrote an old remembered key (... COL3o srey...), not the PC's (... LuhpI0dY chedam-dev)	Re-flash; REMOVE old keys, BROWSE to id_ed25519.pub
A command pasted into Imager's key box	Get-Content ... \ Set-Clipboard text pasted as the key	REMOVE, then BROWSE to the file
"REMOTE HOST IDENTIFICATION HAS CHANGED"	Expected after a re-flash (new host key)	ssh-keygen -R 192.168.50.101
Reservation saved as Disabled (TP-Link)	Router default	Tick it, then Enable Selected; leave ARP binding off
HDMI blank	dmesg: vc4-drm: Cannot find any crtc or sizes	Append video=HDMI-A-1:1280x720@60D to the single line in /boot/firmware/cmdline.txt, then reboot
Lines after sudo cp lost	The sudo password prompt ate the pasted lines	Paste one command at a time when sudo may prompt

HDMI fix commands:

```
sudo cp /boot/firmware/cmdline.txt /boot/firmware/cmdline.txt.bak
sudo sed -i '1 s/$/ video=HDMI-A-1:1280x720@60D/' /boot/firmware/cmdline.txt
sudo reboot
```

Lessons

- Before flashing, check that the key shown in Imager ends the same as type C: \Users\Venkata\.ssh\id_ed25519.pub. Imager remembers old keys.
- Always back up .ssh before changes.
- With password login off, losing id_ed25519 means re-flashing or using a keyboard and screen.
- No passwords or key contents are stored in the repo.

P0 kickoff: spec review, dev PC tools, first Pi flash and first hub setup (TELUS network)

Summary

- Read Master Specification v1.0 and pulled out the P0 (Foundation) scope, done-criteria and phase gate.
- Wrote docs/P0-readiness-checklist.md: hardware, software with versions and links, Pi flashing steps, accounts.
- Installed and verified the developer tools on the Windows PC (all checksums matched the official ones).
- Flashed the Pi. The first attempts failed on SSH keys (including the private key being pasted into Imager); the key was replaced and SSH worked.
- Wrote the hub setup (hub/scripts/setup-hub.sh + hub/system/*) and ran it on the Pi at 192.168.1.241 (TELUS network). HTTPS worked by name and IP.
- Wrote docs/P0-mini-spec.md (P0 build order, decisions DL-21 to DL-25).
- Decisions from Sreya: RTC skipped for now; cloud backup tested later when client IDs exist; test phones are Android 9 and iPhone 13 Pro (iOS 26.6.1); update channel option A (source stays private, signed packages in a separate public repo).

1. Spec review

Source: SDLC Docs Toosl software installers and ReadMe/MASTER Specifications/Chedam - Master Specification v1.0.pdf (41 pages).

The PDF text was extracted with pdftotext -layout. The release-plan and architecture pages are images, so they were rendered with PyMuPDF and read as pictures.

P0 requirements: FR-1.01 to 1.10, 1.13, 1.14, 1.15; FR-12.01 to 12.04, 12.08, 12.09.

Done when: 3 devices log in, the hub survives a power pull, an update installs and rolls back. The phase gate adds: memory under 150 MB (NFR-08), a load test (R6), an iOS/Android HTTPS camera test (R5), and a verified backup and restore.

2. Developer PC tools

Already present: Git 2.52.0, VS Code 1.135, Python 3.13, Windows OpenSSH.

Tool	Version	Verified against	Result
Node.js	24.21.0 (LTS Krypton)	nodejs.org SHASUMS256 + OpenJS signature	Match, installed
PocketBase (Windows)	0.40.4	PocketBase checksums.txt	Match
DuckDB CLI	1.5.6	GitHub release digest	Match
Raspberry Pi Imager	2.0.11.1	GitHub digest + Raspberry Pi Ltd signature	Match, installed
DB Browser for SQLite	3.13.1	winget manifest + SignPath signature	Match, installed
mkcert	1.4.4	Scoop manifest (file is unsigned)	Match
GitHub CLI	2.102.0	n/a	Installed, logged in as repatikosamid-prog

Commands used for verification (Git Bash):

```
sha256sum *.msi *.exe *.zip
curl -s https://nodejs.org/dist/v24.21.0/SHASUMS256.txt | grep x64.msi
```

```
powershell Get-AuthenticodeSignature <file>
```

Failed case: `mkcert` not recognized

- Symptom: `mkcert -install` → "The term 'mkcert' is not recognized".
- Cause: the file was still named `mkcert-v1.4.4-windows-amd64.exe`, and `C:\Tools` was not on PATH. PowerShell also never runs a program from the current folder without `.\`.
- Fix: copied `mkcert.exe`, `pocketbase.exe` and `duckdb.exe` to the top of `C:\Tools` and added `C:\Tools` to the user PATH. A new PowerShell window was needed for the change to show.

Then Sreya ran `mkcert -install` (local dev CA added to the Windows trust store) and `gh auth login` (fine-grained token).

The installers folder was added to `.gitignore`. PocketBase had created test data there.

3. Flashing the Pi (first time)

Imager settings: Pi Zero 2 W, Raspberry Pi OS Lite 64-bit, hostname `chedam`, user `chedam`, Wi-Fi 2.4 GHz, country CA, time zone America/Vancouver, SSH public-key only.

Failed case 1: hostname and username

- The hostname was typed as `Chedam` and the username as the GitHub name.
- Fix: lowercase `chedam` for both, so every script and guide matches.

Failed case 2: no SSH key existed

- SSH was set to password authentication, and the PC had no key yet.
- Fix: `ssh-keygen -t ed25519 -C "chedam-dev"`.

Failed case 3: Permission denied (publickey)

- Debug output showed the PC offering its key and the Pi refusing it for both usernames:

```
ssh -v -o BatchMode=yes chedam@chedam.local true
```

- Cause: the key written to the card was not the PC's key.
- The "host key differs" warning came from an old `192.168.1.241` entry in `known_hosts`. It was removed with `ssh-keygen -R`.

Failed case 4 (security): private key pasted into Imager

- Imager listed 7 "keys". They were the 7 lines of the **private** key (`BEGIN OPENSSH PRIVATE KEY`).
- Risk: the private key was copied onto the SD card and partly shown in a screenshot.
- Fix: removed all 7 entries, deleted the key pair, generated a new one, and pasted **only** `id_ed25519.pub` (one line, `ssh-ed25519 ... chedam-dev`). Re-flashing overwrote the card.

Failed case 5: `.local` name not resolving from Windows tools

- Git Bash ssh and Windows ssh both failed on `chedam.local`, while `Resolve-DnsName` worked.
- Fix: an SSH alias `chedam` pointing at the IP, in `~/.ssh/config`. The Pi first booted slowly (~5 min), so early checks also failed.

Login worked: `chedam@chedam`, Debian 13 trixie, aarch64, 415 MB RAM, 117 GB card.

Admin access

`sudo` asked for a password, so Sreya ran this once:

```
ssh -t chedam "echo 'chedam ALL=(ALL) NOPASSWD:ALL' | sudo tee /etc/sudoers.d/010-chedam && sudo chmo
```

4. Hub setup (first time, Pi at 192.168.1.241)

OS update (ran in the background, ~15 min):

```
sudo apt-get update && sudo apt-get -y full-upgrade && sudo apt-get -y install chrony sqlite3 ufw
```

PocketBase 0.40.4 linux_arm64 was downloaded on the Pi and checked with `sha256sum -c` against `checksums.txt`.

Files written in the repo:

File	Purpose
hub/scripts/setup-hub.sh	Idempotent setup: packages, service user <code>chedam-hub</code> , <code>/opt/chedam</code> , PocketBase service, Caddy, watchdog, journald, Wi-Fi power save off, boot config, firewall
hub/system/chedam-hub.service	PocketBase on 127.0.0.1:8090, auto-restart, <code>GOMEMLIMIT=100MiB</code> , <code>MemoryMax=220M</code> , hardening
hub/system/Caddyfile	HTTPS for <code>chedam.local</code> and the hub IP, using a per-store CA; plain HTTP only for <code>/ca.crt</code>
hub/system/watchdog.conf	Hardware watchdog 14 s
hub/system/journald.conf	Logs capped at 64 MB (SD card wear)
hub/system/wifi-powersave-off.conf	Lower Wi-Fi latency
hub/system/index.html	Placeholder page "Chedam hub is running"

Run on the Pi:

```
scp -r scripts system chedam:hub/
ssh chedam 'sudo bash ~/hub/scripts/setup-hub.sh'
ssh chedam 'sudo systemctl reboot'
```

Failed case 6: GitHub repo creation

- `gh repo create ... --public` → "Resource not accessible by personal access token".
- Cause: the fine-grained token cannot create repositories. Sreya has to create the public repo on github.com.

Failed case 7: HTTPS tests from the PC returned nothing

- Windows `curl` (Schannel) → "the revocation status is unknown". A private CA has no revocation server.
- Fix for command-line tests: `--ssl-no-revoke`. Browsers are not affected.

Failed case 8: Caddy error at start

- Caddy tried to install its root certificate into the Pi's own trust store and failed ("failed to execute sudo").
- Fix: the `skip_install_trust` global option in the Caddyfile.

Results after reboot

Check	Result
Services	chedam-hub, caddy, chrony, avahi-daemon active
HTTPS	<code>https://chedam.local/api/health</code> and <code>https://192.168.1.241/api/health</code> → 200
Certificate	Valid against "Chedam Store CA - 2026 ECC Root" (valid to 2036); IP cert has IP SAN
Memory	PocketBase 27 MB + Caddy 41 MB; total RAM rose from 415 to 462 MB after headless config
Watchdog	14 s
Temperature	52.6 °C, no throttling

5. Decisions recorded (docs/P0-mini-spec.md)

ID	Decision
DL-21	HTTPS via Caddy with an internal CA created on each hub (one per store); mkcert is dev-PC only
DL-22	Source repo private; signed update packages in a separate public repo
DL-23	RTC deferred for dev; required for the pilot
DL-24	Headless boot config (later changed to a switch, see 2026-10-05)
DL-25	PocketBase extended with JavaScript hooks; no Go build

Findings:

- Android 9 cannot resolve `.local`, so the hub needs a fixed IP.
- Android 9's Chrome no longer gets updates.
- Caddy leaf certificates last 12 hours, so the clock must be correct (RTC for the pilot).